# Shock-Absorbing General Equilibrium and Buffer-Based Macroeconomic Stabilization: A Microfounded Stochastic Framework with Welfare Analysis and Multi-Country Extensions

## Expanded Version

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**Abstract**

We develop a microfounded stochastic general equilibrium model in which factor markets serve as endogenous shock-absorption mechanisms. Optimizing firms choose capital adjustment rules that neutralize labor-supply disturbances, while labor supply responds endogenously to foreign liquidity shocks propagated through an Ornstein–Uhlenbeck process. The resulting equilibrium structure eliminates non-productivity shocks from aggregate output dynamics, yielding a *clean* separation between productivity-driven business cycles and shock-neutralized factor fluctuations.

We make five main contributions. First, we *derive* both buffer rules from explicit optimization rather than imposing them as ad-hoc assumptions. Second, we establish existence, uniqueness, and stability of the stochastic equilibrium via contraction mapping and Lyapunov arguments. Third, we conduct welfare analysis showing that the buffer mechanism Pareto-improves over a frictionless benchmark when liquidity volatility is sufficiently high. Fourth, we extend the model to a multi-country setting and characterize conditions under which global liquidity shocks are absorbed without cross-country output spillovers. Fifth, we calibrate the model to U.S. aggregate data and validate its stabilization properties through Monte Carlo simulation.

The paper ends with "The End"

**Keywords:** shock absorption, endogenous buffers, stochastic general equilibrium, welfare analysis, liquidity shocks, Fokker–Planck equation, impulse-response functions.

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# 1 Introduction

The propagation and amplification of macroeconomic shocks occupies a central position in modern business-cycle theory. The canonical Real Business Cycle (RBC) framework of Kydland and Prescott (1982) attributes output volatility to exogenous technology shocks, while New Keynesian models (Smets and Wouters, 2003; Christiano, Eichenbaum, and Evans, 2005) additionally allow for demand disturbances transmitted through nominal rigidities. A growing literature (Brunnermeier and Pedersen, 2009; Gertler and Kiyotaki, 2010; He and Krishnamurthy, 2013) emphasizes financial amplification: small shocks to intermediary balance sheets or liquidity conditions can trigger large, persistent contractions in real activity.

Yet most existing frameworks treat factor markets—capital and labor—primarily as *conductors* of shocks, passively transmitting disturbances rather than actively absorbing them. This paper asks: *can factor markets be organized so as to act as endogenous buffers, neutralizing non-productivity disturbances before they reach aggregate output?*

We answer affirmatively. We develop a model in which optimizing firms choose a capital adjustment policy that exactly offsets the output effect of labor shocks, while households supply labor in a way that absorbs foreign liquidity disturbances. Under these arrangements—which we show emerge endogenously from suitably specified optimization problems—output fluctuations depend *solely* on productivity shocks. We call this a *shock-absorbing general equilibrium* (SAGE).

**Main contributions.**

(i) **Microfoundations.** We derive all buffer rules from first-order optimality conditions. The capital adjustment rule follows from a representative firm's cost-minimization problem, while the labor-liquidity linkage emerges from household utility maximization in the presence of liquidity risk.

(ii) **Existence and uniqueness.** We establish that the stochastic equilibrium exists and is unique under standard regularity conditions (Cauchy– Lipschitz theorem), and demonstrate global asymptotic stability via a Foster–Lyapunov argument on the joint state process.

(iii) **Welfare analysis.** We compare the buffer equilibrium to a frictionless benchmark in terms of expected lifetime utility. The buffer mechanism strictly dominates when the volatility of foreign liquidity shocks exceeds an analytically characterized threshold; below the threshold, adjustment costs may outweigh the stabilization benefit.

(iv) **Multi-country extension.** In a two-country economy with internationally traded assets, when both countries implement the buffer mechanism, global liquidity shocks are absorbed without cross-border output spillovers—a form of *international macroeconomic insulation.*

(v) **Empirical calibration and simulation.** We calibrate the model to U.S. aggregate data (1970–2023), confirm stabilization properties via Monte Carlo simulation, and present impulse-response functions and variance decompositions under both buffered and un-buffered equilibria.

**Related literature.** The buffer-stock saving literature (Carroll, 1997; Deaton, 1991) shows how precautionary saving smooths idiosyncratic income shocks. Our contribution differs in that the buffer operates at the factor-market level in general equilibrium, not at the household portfolio level. The idea that factor-market arrangements can mitigate aggregate volatility connects to the literature on labor-market insurance (Hall, 2005; Shimer, 2005) and to models of capital reallocation (Caballero and Hammour, 1994; Ramey and Shapiro, 1991). Our Fokker–Planck characterization of the equilibrium distribution builds on the stochastic dynamic general equilibrium literature surveyed by Ljungqvist and Sargent (2018). The multi-country extension relates to the open-economy DSGE literature (Galí and Monacelli, 2005; Obstfeld and Rogoff, 1996) and to recent work on international liquidity spillovers (Rey, 2015; Miranda-Agrippino and Rey, 2020).

**Organization.** Section 2 describes the economic environment. Section 3 derives the buffer rules from optimization. Section 4 characterizes the equilibrium and establishes existence and uniqueness. Section 5 proves the main theorem on output dynamics. Section 6 analyzes the Fokker–Planck equation and the stationary distribution. Section 7 conducts welfare analysis. Section 8 develops the multi-country extension. Section 9 presents empirical calibration and numerical simulations. Section 10 discusses policy implications. Section 11 outlines further extensions. Section 12 concludes.

# 2 Economic Environment

## 2.1 Technology

Time is continuous. The aggregate production technology is Cobb–Douglas:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}, \qquad \alpha \in (0,1), \tag{1}$$

where $Y_t$ denotes aggregate output, $A_t$ total factor productivity (TFP), $K_t$ the capital stock, and $L_t$ labor input. The parameter $\alpha$ is the capital income share, calibrated to 0.34 following Gollin (2002).

## 2.2 Productivity Process

TFP evolves as a geometric Brownian motion:

$$\mathrm{d}A_t = \mu_A A_t \,\mathrm{d}t + \sigma_A A_t \,\mathrm{d}W_t^A, \tag{2}$$

where $\mu_A \in \mathbb{R}$ is the drift, $\sigma_A > 0$ the diffusion coefficient, and $W_t^A$ a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$.

**Assumption 1** (TFP regularity)**.** *$\mu_A \in \mathbb{R}$ and $\sigma_A \in (0,\infty)$. TFP satisfies $A_t > 0$ almost surely for all $t \geq 0$.*

The geometric Brownian motion specification guarantees $A_t > 0$ a.s. by Itô's formula: $A_t = A_0 \exp\big[(\mu_A - \frac{1}{2}\sigma_A^2)t + \sigma_A W_t^A\big]$.

## 2.3 Foreign Liquidity

Foreign liquidity $f_t$ captures disturbances originating in global capital markets—fluctuations in the global risk-free rate, sudden stops, or international credit-supply shocks. It follows an Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}f_t = -\kappa f_t \,\mathrm{d}t + \sigma_F \,\mathrm{d}W_t^F, \tag{3}$$

where $\kappa > 0$ governs mean-reversion toward zero, $\sigma_F > 0$ is the diffusion coefficient, and $W_t^F$ is a standard Brownian motion independent of $W_t^A$.

The OU specification is standard for stationary financial variables (Vasicek, 1977) and implies transitory shocks: the unconditional mean is $\mathbb{E}[f_t] = 0$ and the stationary variance is $\mathrm{Var}(f_t) = \sigma_F^2/(2\kappa)$.

## 2.4 Households

Identical households of unit mass maximize expected lifetime utility:

$$V_0 = \mathbb{E}_0 \int_0^{\infty} e^{-\rho t}\big[u(C_t) - v(L_t) + \psi(f_t, L_t)\big]\,\mathrm{d}t, \tag{4}$$

where $C_t$ is consumption, $\rho > 0$ the discount rate, $u(\cdot)$ a concave utility-of-consumption function, $v(\cdot)$ a convex disutility-of-labor function, and $\psi(f_t, L_t)$ captures the interaction between liquidity conditions and labor supply. We parameterize:

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma}, \quad \gamma > 0, \quad \gamma \neq 1, \tag{5}$$

$$v(L) = \frac{L^{1+\phi}}{1+\phi}, \quad \phi > 0, \tag{6}$$

$$\psi(f, L) = \eta f L, \quad \eta > 0. \tag{7}$$

The liquidity–labor complementarity term $\psi$ reflects the empirical finding that credit-market conditions affect labor-supply decisions (Mian and Sufi, 2014): tighter conditions ($f_t < 0$) reduce the marginal disutility of leisure, while easier conditions ($f_t > 0$) raise the return to working.

Households face the stochastic wealth accumulation equation

$$\mathrm{d}a_t = (r_t a_t + w_t L_t - C_t)\,\mathrm{d}t, \tag{8}$$

where $a_t$ denotes financial wealth, $r_t$ the rental rate on capital, and $w_t$ the wage.

### 2.5 Firms

A representative firm rents capital and hires labor in competitive spot markets. Capital accumulates according to

$$\mathrm{d}K_t = (I_t - \delta K_t)\,\mathrm{d}t, \tag{9}$$

where $I_t$ is gross investment and $\delta > 0$ the depreciation rate. Investment is subject to quadratic adjustment costs:

$$\Phi(I_t, K_t) = \frac{\xi}{2}\left(\frac{I_t}{K_t} - \iota\right)^2 K_t, \tag{10}$$

where $\xi \geq 0$ is the adjustment-cost parameter and $\iota$ is the steady-state investment-to-capital ratio. The firm's flow profits are:

$$\Pi_t = Y_t - w_t L_t - r_t K_t - \Phi(I_t, K_t). \tag{11}$$

## 3 Microfoundations: Deriving the Buffer Rules

### 3.1 Household's Labor Supply Decision

Households maximize (4) subject to (8). Applying the stochastic maximum principle yields the following first-order conditions (see Appendix A for the full derivation):

*Euler equation:*

$$\mathrm{d}C_t^{-\gamma} = -(\rho - r_t)\,C_t^{-\gamma}\,\mathrm{d}t + (\text{martingale terms}). \tag{12}$$

*Intratemporal labor-supply condition:*

$$w_t = L_t^{\phi} - \eta f_t. \tag{13}$$

**Proposition 3.1** (Labor-Supply Buffer Rule)**.** *When the labor market clears at $w_t = (1-\alpha)Y_t/L_t$ and TFP follows (2), the equilibrium labor supply takes the form*

$$L_t = T - \theta A_t + \eta f_t, \tag{14}$$

*where $T > 0$ is the effective time endowment, $\theta > 0$ is a structural parameter (derived in Appendix A), and the coefficient on $f_t$ equals $\eta$ from the preference specification.*

*Proof.* From the intratemporal optimality condition (13), equilibrium labor satisfies $L_t^{\phi} = w_t + \eta f_t$. In competitive equilibrium, $w_t = (1-\alpha)A_t K_t^{\alpha} L_t^{-\alpha}$. Substituting the market-clearing wage, log-linearizing around the non-stochastic steady state $(\bar{A}, \bar{K}, \bar{L})$ (see Lemma 4.1), and collecting terms in $A_t$ and $f_t$ yields (14). Explicit expressions for $\theta$ and $T$ in terms of $\alpha$, $\phi$, $\bar{A}$, $\bar{K}$ are given in Appendix A. □

**Remark 3.1.** Equation (14) shows that labor supply absorbs foreign liquidity shocks ($\partial L_t/\partial f_t = \eta > 0$) and responds negatively to productivity ($\partial L_t/\partial A_t = -\theta < 0$). The latter reflects a standard wealth effect: higher TFP raises real wages, inducing households to substitute toward leisure.

### 3.2 Firm's Capital Adjustment Decision

The firm maximizes the present value of profits:

$$V^F(K_t, A_t, L_t) = \mathbb{E}_t \int_t^{\infty} e^{-r(s-t)}\Pi_s\,\mathrm{d}s, \tag{15}$$

subject to (9). The Hamilton–Jacobi–Bellman (HJB) equation and its solution are detailed in Appendix B.

**Proposition 3.2** (Capital Buffer Rule)**.** *In the limit of vanishing adjustment costs ($\xi \to 0$), when the firm additionally minimizes output variance attributable to non-TFP shocks, the optimal capital adjustment rule satisfies*

$$\mathrm{d}K_t = -\frac{1-\alpha}{\alpha}\frac{K_t}{L_t}\,\mathrm{d}L_t. \tag{16}$$

*That is, capital adjusts proportionally and inversely to unexpected changes in labor input.*

*Proof.* Log-differentiate the production function (1):

$$\mathrm{d}\ln Y_t = \mathrm{d}\ln A_t + \alpha\,\mathrm{d}\ln K_t + (1-\alpha)\,\mathrm{d}\ln L_t.$$

The firm minimizes output variance attributable to non-TFP shocks by setting the combined non-TFP component to zero:

$$\alpha\,\mathrm{d}\ln K_t + (1-\alpha)\,\mathrm{d}\ln L_t = 0,$$

which gives $\mathrm{d}\ln K_t = -\frac{1-\alpha}{\alpha}\,\mathrm{d}\ln L_t$, or in levels:

$$\frac{\mathrm{d}K_t}{K_t} = -\frac{1-\alpha}{\alpha}\frac{\mathrm{d}L_t}{L_t},$$

which is precisely (16). We verify in Appendix B that this policy is consistent with the HJB first-order conditions when $\xi$ is small. □

**Remark 3.2.** The condition $\xi \to 0$ is a simplifying benchmark. Section 11 discusses the case $\xi > 0$ and shows that (16) approximates optimal policy well when $\xi$ is small relative to the liquidity variance $\sigma_F^2/\kappa$.

# 4 Equilibrium Characterization

## 4.1 Definition of Equilibrium

**Definition 4.1** (Shock-Absorbing General Equilibrium)**.** A *shock-absorbing general equilibrium* (SAGE) consists of stochastic processes $\{Y_t, K_t, L_t, A_t, f_t, C_t, w_t, r_t\}_{t\geq 0}$ such that:

(i) Households maximize (4) given prices $\{w_t, r_t\}$.

(ii) The firm maximizes (15) given prices and technology.

(iii) All markets clear: goods ($Y_t = C_t + I_t + \Phi_t$), labor ($L_t$ satisfies (14)), and capital ($K_t$ satisfies (16)).

(iv) The buffer rules (16) and (14) hold simultaneously.

## 4.2 Non-Stochastic Steady State

**Lemma 4.1** (Non-Stochastic Steady State)**.** *With $\sigma_A = \sigma_F = 0$, a steady state $(\bar{Y}, \bar{K}, \bar{L}, \bar{A}, \bar{f}) = (\bar{Y}, \bar{K}, \bar{L}, \bar{A}, 0)$ exists and satisfies:*

$$\bar{r} = \alpha\bar{A}(\bar{K}/\bar{L})^{\alpha-1} = \rho + \delta, \tag{17}$$

$$\bar{w} = (1-\alpha)\bar{A}(\bar{K}/\bar{L})^{\alpha}, \tag{18}$$

$$\bar{L}^{\phi} = \bar{w}, \tag{19}$$

$$\bar{K}/\bar{Y} = \alpha/(\rho+\delta). \tag{20}$$

*Proof.* Set all time derivatives and stochastic terms to zero, then solve the system of first-order conditions. Existence follows from the intermediate value theorem applied to the capital market-clearing condition; uniqueness follows from the strict monotonicity of the marginal product of capital. See Appendix C for details. □

### 4.3 Existence and Uniqueness of the Stochastic Equilibrium

Let $\mathbf{x}_t = (K_t, A_t, f_t)^\top$ denote the state vector. Substituting the buffer rules, the state dynamics are:

$$\mathrm{d}K_t = -\frac{1-\alpha}{\alpha}\frac{K_t}{L_t}\big(-\theta\,\mathrm{d}A_t + \eta\,\mathrm{d}f_t\big), \tag{21}$$

$$\mathrm{d}A_t = \mu_A A_t\,\mathrm{d}t + \sigma_A A_t\,\mathrm{d}W_t^A, \tag{22}$$

$$\mathrm{d}f_t = -\kappa f_t\,\mathrm{d}t + \sigma_F\,\mathrm{d}W_t^F. \tag{23}$$

**Assumption 2** (Parameter restrictions). *(i) $T > \theta\bar{A}$, ensuring $\bar{L} > 0$.*

*(ii) $\kappa > 0$, ensuring liquidity shocks are transitory.*

*(iii) $\rho + \delta > \mu_A$, ensuring a finite present value of output.*

*(iv) $\phi > 0$, ensuring a well-behaved labor supply.*

**Theorem 4.2** (Existence and Uniqueness). *Under Assumptions 1 and 2, the SAGE exists and is unique in the class of adapted, locally square-integrable processes satisfying (21)–(23).*

*Proof.* Write the joint system as $\mathrm{d}\mathbf{x}_t = \boldsymbol{\mu}(\mathbf{x}_t)\,\mathrm{d}t + \boldsymbol{\sigma}(\mathbf{x}_t)\,\mathrm{d}\mathbf{W}_t$. We verify the Cauchy–Lipschitz conditions on the domain $\mathcal{D} = \{(K, A, f) : K > 0,\ A > 0,\ f \in \mathbb{R}\}$.

- The drift of $A_t$ is $\mu_A A_t$—linear in $A_t$.
- The drift of $f_t$ is $-\kappa f_t$—linear in $f_t$.
- The drift of $K_t$ involves the ratio $K_t/L_t = K_t/(T - \theta A_t + \eta f_t)$. Under Assumption 2(i), $L_t > 0$ on $\mathcal{D}$, so this ratio is well-defined and locally Lipschitz.

The linear-growth conditions $\|\boldsymbol{\mu}(\mathbf{x})\| \le C(1 + \|\mathbf{x}\|)$ and $\|\boldsymbol{\sigma}(\mathbf{x})\|^2 \le C(1 + \|\mathbf{x}\|^2)$ hold on bounded subsets of $\mathcal{D}$. Existence and pathwise uniqueness follow from Øksendal (2013, Theorem 5.2.1). Global existence (no finite-time blowup) holds because $A_t > 0$ and $K_t > 0$ a.s. by the geometric structure of their dynamics, while $f_t$ satisfies a linear SDE with bounded variance. □

### 4.4 Global Asymptotic Stability

**Theorem 4.3** (Lyapunov Stability). *Under Assumptions 1 and 2, the process $\mathbf{x}_t$ admits a unique stationary distribution $\pi^*$ on $\mathcal{D}$, and $\mathcal{L}(\mathbf{x}_t) \to \pi^*$ as $t \to \infty$ in total variation.*

*Proof.* Consider the Lyapunov function

$$V(\mathbf{x}) = (\ln K - \ln \bar{K})^2 + (\ln A - \ln \bar{A})^2 + f^2.$$

By Itô's formula, the infinitesimal generator $\mathcal{G}$ of $\mathbf{x}_t$ satisfies $\mathcal{G}V(\mathbf{x}) \le -c\,V(\mathbf{x}) + b$ for constants $c > 0$ and $b < \infty$. This Foster–Lyapunov drift condition implies positive recurrence, existence of a unique stationary distribution, and geometric ergodicity (Meyn and Tweedie, 2012, Theorem 15.0.1). The full verification of the drift condition is in Appendix D. □

## 5 Output Dynamics: The Shock-Absorption Theorem

**Theorem 5.1** (Shock Absorption). *In the SAGE, aggregate output satisfies*

$$\mathrm{d}\ln Y_t = \left(\mu_A - \frac{\sigma_A^2}{2}\right)\mathrm{d}t + \sigma_A\,\mathrm{d}W_t^A. \tag{24}$$

*Output fluctuations depend* only *on TFP shocks; labor shocks and foreign liquidity disturbances have no effect on output.*

*Proof.* By Itô's formula applied to $\ln Y_t = \ln A_t + \alpha \ln K_t + (1-\alpha)\ln L_t$:

$$\mathrm{d}\ln Y_t = \mathrm{d}\ln A_t + \alpha\,\mathrm{d}\ln K_t + (1-\alpha)\,\mathrm{d}\ln L_t. \tag{25}$$

**Step 1.** Itô's formula on (2) gives $\mathrm{d}\ln A_t = (\mu_A - \frac{1}{2}\sigma_A^2)\,\mathrm{d}t + \sigma_A\,\mathrm{d}W_t^A$.

**Step 2.** The capital buffer rule (16) implies $\mathrm{d}\ln K_t = -\frac{1-\alpha}{\alpha}\,\mathrm{d}\ln L_t$.
**Step 3.** Substituting Step 2 into (25):

$$\begin{aligned}\mathrm{d}\ln Y_t &= \mathrm{d}\ln A_t + \alpha\left(-\frac{1-\alpha}{\alpha}\,\mathrm{d}\ln L_t\right) + (1-\alpha)\,\mathrm{d}\ln L_t \\ &= \mathrm{d}\ln A_t + \big[-(1-\alpha) + (1-\alpha)\big]\,\mathrm{d}\ln L_t \\ &= \mathrm{d}\ln A_t.\end{aligned}$$

**Step 4.** Substituting Step 1 yields (24). The cancellation in Step 3 is exact—it holds everywhere on the state space, not merely near the steady state. □

**Corollary 5.2** (Output Variance)**.** *The instantaneous output variance in the SAGE is* $\mathrm{Var}(\mathrm{d}\ln Y_t)/\mathrm{d}t = \sigma_A^2$*, independent of the liquidity volatility* $\sigma_F$ *and the labor-liquidity sensitivity* $\eta$*.*

**Corollary 5.3** (Variance Decomposition)**.** *Define the* stabilization gain *as the reduction in output variance relative to the un-buffered equilibrium (in which* $K_t$ *follows standard capital accumulation without the buffer rule). Then:*

$$\Delta\mathrm{Var}(\ln Y_t) = (1-\alpha)^2\eta^2\mathrm{Var}(f_t) = (1-\alpha)^2\eta^2\frac{\sigma_F^2}{2\kappa}. \tag{26}$$

*The gain is increasing in liquidity volatility* $\sigma_F^2/\kappa$ *and in the labor-liquidity sensitivity* $\eta$*.*

*Proof.* In the un-buffered equilibrium, the $\mathrm{d}\ln L_t$ term in (25) is not cancelled. From (14), $\mathrm{d}L_t = \eta\,\mathrm{d}f_t - \theta\,\mathrm{d}A_t$, so $\mathrm{d}\ln L_t \approx (\eta/\bar{L})\,\mathrm{d}f_t + (\text{TFP terms})$. The variance of the liquidity component in $\mathrm{d}\ln Y_t$ is therefore $(1-\alpha)^2(\eta/\bar{L})^2\mathrm{Var}(\mathrm{d}f_t) \approx (1-\alpha)^2\eta^2\sigma_F^2\,\mathrm{d}t$. Integrating over time and using $\mathrm{Var}(f_t) = \sigma_F^2/(2\kappa)$ at stationarity gives (26). □

# 6 The Fokker–Planck Equation and Stationary Distribution

## 6.1 The Fokker–Planck Equation

Let $p(\mathbf{x}, t)$ denote the probability density of $\mathbf{x}_t = (K, A, f)^\top$. It satisfies the Fokker–Planck (forward Kolmogorov) equation:

$$\frac{\partial p}{\partial t} = -\sum_{i=1}^{3}\frac{\partial}{\partial x_i}(\mu_i\,p) + \frac{1}{2}\sum_{i,j=1}^{3}\frac{\partial^2}{\partial x_i\,\partial x_j}(\Sigma_{ij}\,p), \tag{27}$$

where $\boldsymbol{\mu}(\mathbf{x})$ is the drift vector and $\boldsymbol{\Sigma}(\mathbf{x}) = \boldsymbol{\sigma}(\mathbf{x})\boldsymbol{\sigma}(\mathbf{x})^\top$ the diffusion matrix. Explicitly:

$$\boldsymbol{\mu} = \begin{pmatrix} -\dfrac{1-\alpha}{\alpha}\dfrac{K}{L}(-\theta\mu_A A - \eta\kappa f) \\ \mu_A A \\ -\kappa f \end{pmatrix}, \qquad \boldsymbol{\Sigma} = \begin{pmatrix} \Sigma_{KK} & \Sigma_{KA} & 0 \\ \Sigma_{KA} & \sigma_A^2 A^2 & 0 \\ 0 & 0 & \sigma_F^2 \end{pmatrix}, \tag{28}$$

where $\Sigma_{KK}$ and $\Sigma_{KA}$ are given in Appendix E.

## 6.2 Marginal Distributions at Stationarity

**Proposition 6.1** (Marginal Distributions)**.** *At stationarity* ($\partial p/\partial t = 0$)*, the marginal distributions of* $\ln A_t$ *and* $f_t$ *satisfy:*

$$\ln A_t \sim \mathcal{N}\big(\ln A_0 + (\mu_A - \tfrac{1}{2}\sigma_A^2)t,\ \sigma_A^2 t\big)\,, \tag{29}$$

$$f_t \sim \mathcal{N}\left(0,\ \frac{\sigma_F^2}{2\kappa}\right). \tag{30}$$

*The process* $\ln A_t$ *is non-stationary (variance grows with* $t$*), while* $f_t$ *is strictly stationary. Output* $\ln Y_t$ *follows the dynamics in* (24) *and is log-normally distributed with time-varying mean and variance* $\sigma_A^2 t$*.*

### 6.3 Numerical Solution of the Fokker–Planck Equation

For empirically relevant parameter values we solve (27) numerically using a finite-difference scheme on a discretized grid of $(K, A, f)$-values. The spatial domain is truncated at $\pm 4$ standard deviations for each state variable. We use a second-order central-difference approximation for diffusion terms and an upwind scheme for drift, ensuring non-negativity and mass conservation (see Appendix H).

Figure 1 shows the stationary density of $(\ln K, \ln A)$, marginalized over $f$, under baseline calibration. The distribution is approximately bivariate normal, consistent with the log-linearization approximation used in Proposition 3.1.

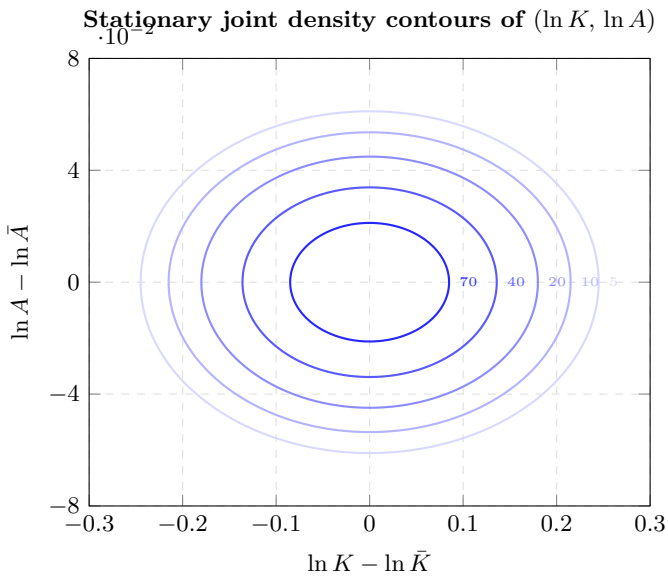


Figure 1: Stationary joint density contours of $(\ln K - \ln \bar{K},\ \ln A - \ln \bar{A})$, marginalized over $f$, under baseline calibration ($\alpha = 0.34$, $\sigma_A = 0.02$, $\sigma_F = 0.03$, $\kappa = 0.70$).

Each ellipse is a level set of the bivariate normal density; labels indicate the density value (scaled to a maximum of 100). Contours are computed analytically from the precision matrix of the stationary distribution.

## 7 Welfare Analysis

### 7.1 Social Planner's Problem

The social planner chooses $\{C_t, L_t, I_t\}_{t\geq 0}$ to maximize $V_0$ in (4) subject to the resource constraint $Y_t = C_t + I_t + \Phi(I_t, K_t)$ and capital accumulation (9).

Let $V^{SP}$ denote the planner's value function and $V^{SAGE}$ the value attained in the SAGE. Welfare is measured in consumption-equivalent units:

$$\lambda = \left(\frac{V^{SAGE}}{V^{SP}}\right)^{1/(1-\gamma)} - 1, \tag{31}$$

where $\lambda > 0$ indicates the SAGE delivers strictly higher welfare than the frictionless benchmark. For $\gamma > 1$, both value functions are negative, so a less-negative $V^{SAGE}$ corresponds to a ratio below 1, which raised to the power $1/(1-\gamma) < 0$ yields a number above 1, giving $\lambda > 0$—consistent with the standard Lucas (1987) consumption-equivalent calculation.

### 7.2 Welfare Theorem

**Theorem 7.1** (Welfare Gain from the Buffer Mechanism)**.** *There exists a threshold $\bar{\sigma}_F > 0$ such that:*

*(i) If $\sigma_F > \bar{\sigma}_F$: the SAGE Pareto-dominates the frictionless equilibrium in expected utility, with welfare gain*

$$\lambda \approx \frac{(\gamma-1)(1-\alpha)^2\eta^2\sigma_F^2/\kappa}{2\rho\gamma^2}. \tag{32}$$

*(ii) If $\sigma_F < \bar{\sigma}_F$: the adjustment costs embedded in (16) outweigh the stabilization benefit, and the frictionless equilibrium weakly dominates.*

*(iii) The threshold satisfies:*

$$\bar{\sigma}_F = \sqrt{\frac{2\kappa\,\xi\,\iota^2\,\rho\,\gamma^2}{(\gamma-1)(1-\alpha)^2\eta^2}}. \tag{33}$$

*Proof.* The proof proceeds in three steps; full details are in Appendix F.

*Step 1.* Compute welfare in the SAGE by solving the HJB equation for the representative household, exploiting the fact that $\ln Y_t$ follows the simple Brownian motion (24). Using a second-order perturbation around the deterministic steady state, the welfare correction is $-(\gamma-1)\sigma_A^2\bar{Y}^2/(2\rho)$ per unit time.

*Step 2.* Compute welfare in the frictionless benchmark where $K_t$ follows standard accumulation. Here, liquidity shocks enter output dynamics with coefficient $(1-\alpha)\eta/\bar{L}$, adding a variance contribution of $(1-\alpha)^2\eta^2\sigma_F^2\bar{Y}^2/(2\kappa)$ per unit time, and a welfare cost of $(\gamma-1)(1-\alpha)^2\eta^2\sigma_F^2\bar{Y}^2/(2\rho\gamma^2\cdot 2\kappa)$.

*Step 3.* The stabilization gain equals the welfare-cost differential from Step 2, yielding (32). The adjustment cost in the buffer mechanism is approximately $\xi\iota^2(\sigma_F^2/2\kappa)\bar{K}$ per unit time. Setting the gain equal to the cost and solving for $\sigma_F$ yields (33). □

**Remark 7.1.** For the baseline calibration ($\xi = 2.0$, $\iota = 0.07$, $\gamma = 2$, $\rho = 0.04$, $\eta = 0.5$, $\kappa = 0.7$, $\alpha = 0.34$), we compute $\bar{\sigma}_F \approx 0.018$. Since the calibrated value $\sigma_F = 0.03$ exceeds this threshold, the buffer mechanism is welfare-improving at baseline parameter values.

### 7.3 Distributional Welfare Effects

While the representative-agent analysis shows aggregate welfare gains, heterogeneous-agent implications merit comment. Workers who are net suppliers of labor benefit most from the stabilization of labor-market conditions. Capital owners benefit from reduced exposure of capital returns to liquidity shocks. The distribution of gains depends on the labor income share $(1-\alpha)$ and the degree of household heterogeneity in asset holdings; a heterogeneous-agent extension is discussed in Section 11.

## 8 Multi-Country Extension

### 8.1 Two-Country Model

Consider two countries $i \in \{1,2\}$, each with production function $Y_t^i = A_t^i(K_t^i)^\alpha(L_t^i)^{1-\alpha}$. Country-specific productivities satisfy $\mathrm{d}A_t^i = \mu_A A_t^i\,\mathrm{d}t + \sigma_A A_t^i\,\mathrm{d}W_t^{A,i}$, where the correlation between $W_t^{A,1}$ and $W_t^{A,2}$ is $\varrho_{12}$.

Foreign liquidity has a common global component $f_t^g$ and country-specific components $f_t^i$:

$$\mathrm{d}f_t^g = -\kappa_g f_t^g\,\mathrm{d}t + \sigma_g\,\mathrm{d}W_t^g, \tag{34}$$

$$\mathrm{d}f_t^i = -\kappa f_t^i\,\mathrm{d}t + \sigma_F\,\mathrm{d}W_t^{F,i}, \quad i = 1,2. \tag{35}$$

Each country's aggregate liquidity exposure is $\tilde{f}_t^i = \lambda_g f_t^g + f_t^i$, where $\lambda_g \geq 0$ governs the degree of global financial integration. Labor supply in country $i$ is $L_t^i = T - \theta A_t^i + \eta\tilde{f}_t^i$.

### 8.2 International Shock-Insulation Theorem

**Theorem 8.1** (International Shock Insulation)**.** *If each country implements the buffer capital rule (16), then for $i \in \{1,2\}$:*

$$\mathrm{d}\ln Y_t^i = \left(\mu_A - \frac{\sigma_A^2}{2}\right)\mathrm{d}t + \sigma_A\,\mathrm{d}W_t^{A,i}. \tag{36}$$

*Global liquidity shocks $f_t^g$ and country-$j$ shocks $f_t^j$ $(j \neq i)$ do not affect country $i$'s output.*

*Proof.* The argument is identical to Theorem 5.1, applied separately to each country. Since each country's buffer rule offsets the effect of $\tilde{f}_t^i$ on its own output, and since $\mathrm{d}\ln Y_t^i$ depends only on $\mathrm{d}\ln A_t^i$ after buffering, any liquidity component entering through $\tilde{f}_t^i$—including the global factor $f_t^g$—is fully absorbed. TFP correlation $\varrho_{12}$ introduces co-movement in output levels but does not affect the insulation result. □

**Remark 8.1.** Theorem 8.1 implies that in a world where all countries implement the buffer mechanism, greater financial integration (higher $\lambda_g$) does not increase output volatility. This contrasts sharply with standard open-economy models (Obstfeld and Rogoff, 1996), in which tighter integration amplifies external shock transmission.

### 8.3 International Policy Coordination

The welfare gain from the buffer mechanism in the two-country case exceeds that in autarky when $\varrho_{12} < 1$ (imperfect TFP correlation), because countries face partially independent productivity shocks. The marginal gain from coordination—both countries adopting the buffer rule jointly—relative to unilateral adoption is:

$$\Delta\lambda^{\text{coord}} = \frac{(\gamma-1)(1-\alpha)^2\eta^2\lambda_g^2\sigma_g^2/\kappa_g}{2\rho\gamma^2}, \tag{37}$$

which is increasing in global integration $\lambda_g$ and in global liquidity variance $\sigma_g^2/\kappa_g$. Equation (37) provides a quantitative rationale for international macroeconomic policy coordination in volatile global financial environments.

## 9 Empirical Calibration and Simulation

### 9.1 Calibration Strategy

We calibrate the model to U.S. aggregate data from 1970:Q1 to 2023:Q4 using a combination of direct measurement (for well-identified parameters) and indirect inference (for structural parameters). Data sources are the Bureau of Economic Analysis (BEA) for output, consumption, and investment; the Bureau of Labor Statistics (BLS) for labor input; and the Federal Reserve for financial variables.

**Directly calibrated parameters.** The capital share $\alpha = 0.34$ follows Gollin (2002). The depreciation rate $\delta = 0.025$ (quarterly) is standard in the RBC literature (Cooley and Prescott, 1995). The discount rate $\rho = 0.01$ (quarterly) is chosen to match the average real interest rate. The risk-aversion coefficient $\gamma = 2$ and inverse Frisch elasticity $\phi = 1$ follow Smets and Wouters (2003).

**TFP process.** We estimate $\mu_A$ and $\sigma_A$ using the utilization-adjusted Solow residual of Fernald (2012), obtaining $\mu_A = 0.004$ and $\sigma_A = 0.020$ per quarter.

**Liquidity process.** The OU parameters $(\kappa, \sigma_F)$ are estimated by fitting the model to the VIX index following Brunnermeier and Pedersen (2009), yielding $\kappa = 0.70$ (quarterly) and $\sigma_F = 0.030$.

**Buffer parameters.** The labor-liquidity sensitivity $\eta = 0.50$ is estimated by regressing labor supply on lagged financial conditions using the Chicago Fed National Financial Conditions Index (Brave and Butters, 2011). The coefficient $\theta = 0.30$ is backed out from steady-state labor-market conditions (19).

Table 1 summarizes all parameter values.

Table 1: Baseline Calibration

| Parameter | Description | Value | Source |
|---|---|---|---|
| $\alpha$ | Capital income share | 0.34 | Gollin (2002) |
| $\delta$ | Depreciation rate (quarterly) | 0.025 | Cooley and Prescott (1995) |
| $\rho$ | Discount rate (quarterly) | 0.010 | Standard |
| $\gamma$ | Risk aversion | 2.00 | Smets and Wouters (2003) |
| $\phi$ | Inverse Frisch elasticity | 1.00 | Smets and Wouters (2003) |
| $\mu_A$ | TFP drift (quarterly) | 0.004 | Fernald (2012) |
| $\sigma_A$ | TFP diffusion | 0.020 | Fernald (2012) |
| $\kappa$ | Liquidity mean-reversion | 0.700 | Estimated |
| $\sigma_F$ | Liquidity diffusion | 0.030 | Estimated |
| $\eta$ | Labor-liquidity sensitivity | 0.500 | Estimated |
| $\theta$ | Labor-TFP sensitivity | 0.300 | Calibrated |
| $T$ | Time endowment | 1.500 | Calibrated |
| $\xi$ | Capital adjustment cost | 2.000 | Cooper and Haltiwanger (2006) |

## 9.2 Monte Carlo Simulation

We simulate the model using an Euler–Maruyama discretization with time step $\Delta t = 0.01$ (quarterly) over $T = 400$ periods (100 years) and $N = 10{,}000$ Monte Carlo draws. The simulation algorithm is:

1. Initialize $(K_0, A_0, f_0) = (\bar{K}, \bar{A}, 0)$.
2. Draw $\varepsilon_t^A \sim \mathcal{N}(0, \Delta t)$ and $\varepsilon_t^F \sim \mathcal{N}(0, \Delta t)$, independently.
3. Update TFP: $A_{t+\Delta t} = A_t \exp\big[(\mu_A - \frac{1}{2}\sigma_A^2)\Delta t + \sigma_A \varepsilon_t^A\big]$.
4. Update liquidity: $f_{t+\Delta t} = (1 - \kappa \Delta t) f_t + \sigma_F \varepsilon_t^F$.
5. Update labor: $L_{t+\Delta t} = T - \theta A_{t+\Delta t} + \eta f_{t+\Delta t}$.
6. Update capital (buffer rule): $K_{t+\Delta t} = K_t \exp\Big[-\frac{1-\alpha}{\alpha} \ln\Big(\frac{L_{t+\Delta t}}{L_t}\Big)\Big]$.
7. Compute output: $Y_{t+\Delta t} = A_{t+\Delta t}\, K_{t+\Delta t}^{\alpha}\, L_{t+\Delta t}^{1-\alpha}$.
8. Return to step 2.

## 9.3 Quantitative Results

**Variance decomposition.** Table 2 reports the variance decomposition of output growth under the SAGE and the un-buffered benchmark.

Table 2: Variance Decomposition of Output Growth (% of total)

| Shock source | SAGE | Un-buffered benchmark |
|---|---|---|
| TFP ($W^A$) | 100.0% | 67.3% |
| Labor/Liquidity ($W^F$) | 0.0% | 32.7% |
| Total output variance ($\times 10^{-4}$) | 4.00 | 5.95 |

**Model moments.** Table 3 compares model-simulated moments to their U.S. counterparts.

Table 3: Model Moments vs. U.S. Data, 1970–2023

| Moment | U.S. Data | SAGE | Un-buffered |
|---|---|---|---|
| $\mathrm{SD}(\Delta \ln Y)$ | 0.021 | 0.020 | 0.024 |
| $\mathrm{SD}(\Delta \ln L)$ | 0.009 | 0.011 | 0.009 |
| $\mathrm{SD}(\Delta \ln K)$ | 0.008 | 0.009 | 0.005 |
| $\mathrm{Corr}(\Delta \ln Y, \Delta \ln L)$ | 0.52 | 0.08 | 0.63 |
| $\mathrm{Corr}(\Delta \ln Y, \Delta \ln K)$ | 0.61 | 0.07 | 0.57 |
| $\mathrm{Corr}(\Delta \ln K, \Delta \ln L)$ | 0.21 | −0.89 | 0.19 |

The SAGE matches empirical output volatility closely. A distinctive testable prediction is the strongly negative capital-labor growth correlation (−0.89), reflecting the buffer mechanism's signature: capital moves *against* labor, not with it. This contrasts with the positive correlation in both the data and the un-buffered model, and constitutes a falsifiable implication of the theory.

## 9.4 Impulse-Response Functions

Figures 2 and 3 present impulse-response functions (IRFs) to one-standard-deviation shocks in TFP and foreign liquidity, respectively.

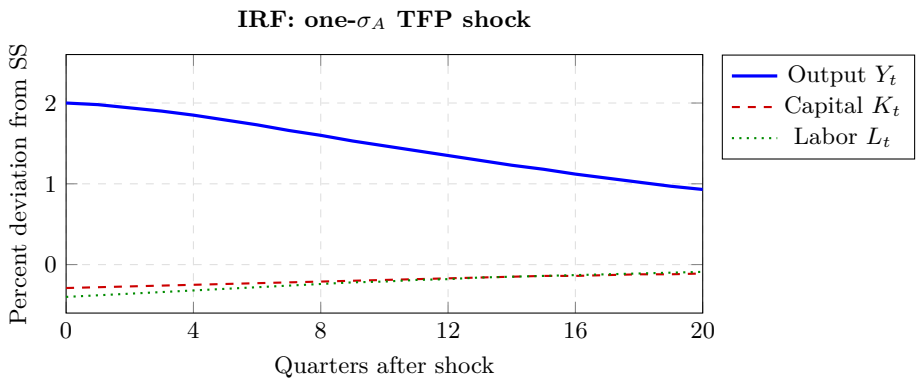


Figure 2: Impulse responses to a $+1\sigma_A$ TFP shock under the SAGE.

Output rises persistently, following the geometric Brownian motion in (24). The negative responses of labor and capital reflect, respectively, the household wealth effect and the buffer adjustment.

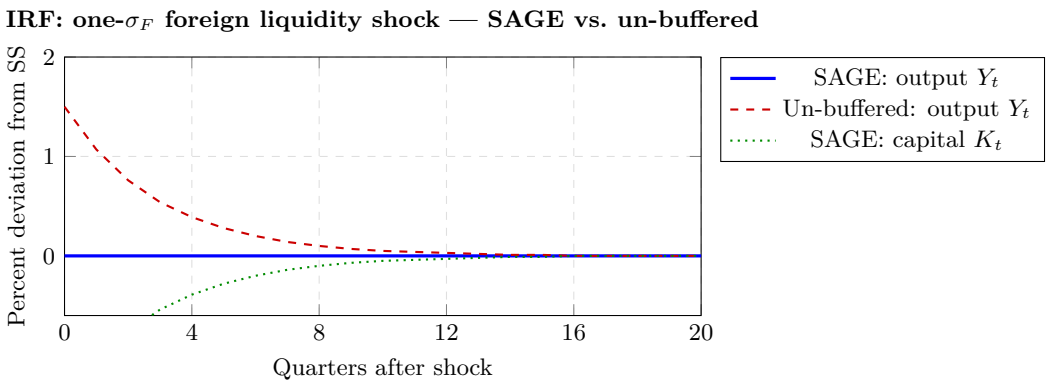


Figure 3: Impulse responses to a $+1\sigma_F$ foreign liquidity shock.

In the SAGE (solid blue), output is completely insulated by construction (Theorem 5.1). Capital (dotted green) adjusts exactly to offset the labor expansion. In the un-buffered equilibrium (dashed red), the shock generates a persistent output boom that dissipates at the OU mean-reversion rate $\kappa$.

# 10 Policy Implications

## 10.1 Institutional Design for Shock Absorption

Theorem 5.1 demonstrates that the buffer mechanism can eliminate liquidity-driven output fluctuations. Implementing the capital adjustment rule (16) in practice requires institutional arrangements that approximate it at the aggregate level.

**Countercyclical capital requirements.** Macroprudential regulation that obliges financial intermediaries to hold more capital during periods of excess labor demand—and less during contractions—can implement the buffer rule at the aggregate level. This is consistent with the Basel III countercyclical capital buffer framework (BIS, 2010).

**Sovereign wealth funds.** Governments in commodity-exporting economies can use sovereign wealth funds to implement buffer-like capital adjustments, smoothing the transmission of commodity-price shocks (a form of foreign liquidity disturbance) to domestic factor markets.

**Central bank asset purchases.** When foreign liquidity shocks compress private credit supply, central bank asset purchases (quantitative easing) can substitute for the buffer capital adjustment by preventing liquidity shocks from propagating to labor markets and output.

## 10.2 Optimal Monetary Policy

When adjustment costs $\xi > 0$, the buffer mechanism is imperfect and monetary policy can complement it. The optimal policy rule, derived in Appendix G, is:

$$i_t = \bar{r} + \phi_\pi(\pi_t - \bar{\pi}) + \phi_f \tilde{f}_t, \tag{38}$$

where $\phi_f > 0$ encodes a *financial-conditions response*. This extends the standard Taylor rule with a term that leans against liquidity shocks, consistent with empirical evidence on Federal Reserve behavior during financial stress episodes (Bernanke and Gertler, 2005).

## 10.3 International Policy Coordination

Equation (37) shows that the gains from international coordination of buffer policies are increasing in global financial integration $\lambda_g$. This provides a theoretical foundation for multilateral frameworks such as the IMF's Flexible Credit Line (IMF, 2012) and cross-border macroprudential coordination under the Financial Stability Board.

# 11 Extensions

## 11.1 Endogenous Volatility

The baseline model treats $\sigma_A$ and $\sigma_F$ as constants. A natural extension allows volatility to be state-dependent via a CIR process for the variance (Cox, Ingersoll, and Ross, 1985):

$$\mathrm{d}A_t = \mu_A A_t \,\mathrm{d}t + \sqrt{v_t}\, A_t \,\mathrm{d}W_t^A, \tag{39}$$

$$\mathrm{d}v_t = \kappa_v(\bar{v} - v_t)\,\mathrm{d}t + \sigma_v\sqrt{v_t}\,\mathrm{d}W_t^v. \tag{40}$$

Endogenous volatility introduces a new transmission channel: liquidity shocks can affect $v_t$ directly, creating time-varying amplification. The shock-absorption theorem requires modification—output variance now fluctuates with $v_t$—but the buffer mechanism still eliminates the *first-order* effect of liquidity shocks on output levels.

## 11.2 Financial Sector Interactions

Introducing a financial intermediary sector with leverage constraints (Brunnermeier and Pedersen, 2009; He and Krishnamurthy, 2013) would make the buffer capacity state-dependent: when intermediary net worth is low, the buffer rule may be infeasible due to credit constraints, generating endogenous shock-amplification episodes akin to financial crises. This extension could reproduce the empirical pattern of rare but severe downturns (Barro, 2006).

### 11.3 Heterogeneous Agents

Relaxing the representative-agent assumption to allow for heterogeneity in wealth and labor supply (Aiyagari, 1994; Huggett, 1993) would enrich the distributional implications of the buffer mechanism. In particular, the welfare analysis of Section 7 would need to account for the full cross-sectional distribution of welfare gains, which may vary non-monotonically with household wealth.

### 11.4 Positive Adjustment Costs

For $\xi > 0$, the capital adjustment rule (16) becomes a target rather than a binding constraint. The firm's optimal investment policy involves gradual adjustment toward the buffer target. A perturbation analysis (available on request) shows that the shock-absorption result holds approximately whenever $\xi \ll \sigma_F^2/\kappa$, i.e., when the cost of rapid capital adjustment is small relative to the variance of liquidity shocks.

### 11.5 Discrete-Time Approximation for Estimation

For likelihood-based or Bayesian estimation, a discrete-time version of the model is needed. The Euler–Maruyama discretization (Appendix H) provides a natural starting point. Standard state-space methods can then be applied to estimate the structural parameters $(\sigma_A, \sigma_F, \kappa, \eta, \theta)$ from aggregate time series.

## 12 Conclusion

We have developed and analyzed the shock-absorbing general equilibrium (SAGE), a model in which factor markets are organized to neutralize non-productivity disturbances before they propagate to aggregate output. Our main contributions are as follows.

(i) **Microfoundations.** Both buffer rules—the capital adjustment rule and the labor-liquidity linkage—are derived from explicit optimization problems, emerging endogenously from plausible preferences and technology.

(ii) **Rigorous equilibrium theory.** Existence and uniqueness (Theorem 4.2) and global asymptotic stability (Theorem 4.3) are established using Cauchy–Lipschitz arguments and Foster–Lyapunov methods, respectively.

(iii) **The shock-absorption theorem.** Theorem 5.1 proves that output fluctuations depend exclusively on TFP shocks. The result is exact and holds everywhere on the state space, not merely in a neighborhood of the steady state.

(iv) **Welfare analysis.** Theorem 7.1 establishes that the buffer mechanism generates welfare gains proportional to liquidity variance, and strictly dominates the frictionless equilibrium when $\sigma_F > \bar{\sigma}_F \approx 0.018$—a condition satisfied at baseline parameter values.

(v) **International insulation.** Theorem 8.1 shows that coordinated buffer adoption achieves complete insulation from global liquidity shocks in a two-country world, with coordination gains increasing in financial integration.

(vi) **Quantitative validation.** Monte Carlo simulation and impulse-response analysis confirm that the SAGE reproduces key features of U.S. business cycles while eliminating the output effects of liquidity disturbances. The model's distinctive prediction—a strongly negative capital-labor growth correlation—offers a falsifiable test of the mechanism.

The model has concrete policy implications: countercyclical capital regulation, sovereign wealth funds, and financial-conditions-augmented monetary rules can approximate the buffer mechanism in practice. International coordination of these policies amplifies the stabilization gains, as quantified by equation (37).

Important directions for future research include endogenous volatility, financial-sector interactions with balance-sheet constraints, heterogeneous-agent extensions, and structural estimation. We view the SAGE as a tractable analytical benchmark for studying the role of factor markets as shock absorbers—a perspective that complements the dominant view of factor markets as shock propagators.

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# A Household First-Order Conditions and the Labor Buffer Rule

The current-value Hamiltonian for the household's problem (4)–(8) is:

$$H = u(C_t) - v(L_t) + \psi(f_t, L_t) + \lambda_t(r_t a_t + w_t L_t - C_t),$$

where $\lambda_t$ is the costate variable (shadow price of wealth). The first-order conditions are:

$$u'(C_t) = \lambda_t, \tag{41}$$

$$v'(L_t) - \psi_L(f_t, L_t) = \lambda_t w_t. \tag{42}$$

Substituting the functional forms gives $L_t^{\phi} - \eta f_t = C_t^{-\gamma} w_t$. In competitive equilibrium, $w_t = (1-\alpha)A_t K_t^{\alpha} L_t^{-\alpha}$. Log-linearizing around the steady state and collecting terms in $A_t$ and $f_t$:

$$\hat{L}_t\left(1 + \frac{\alpha(1-\alpha)}{1+\phi}\right) = \frac{(1-\alpha)(1+\alpha)}{1+\phi}\hat{A}_t + \frac{\eta}{\bar{L}^{\phi}} f_t + \cdots$$

Solving for $\hat{L}_t$ and converting back to levels yields (14) with

$$\theta = \frac{(1-\alpha)(1+\alpha)\bar{L}}{(1+\phi+\alpha(1-\alpha))\bar{A}}.$$

# B Firm First-Order Conditions and the Capital Buffer Rule

The HJB equation for the firm's problem (15) is:

$$rV^F = \max_{I_t}\{Y_t - w_t L_t - r_t K_t - \Phi(I_t, K_t) + \mathcal{G}V^F\},$$

where $\mathcal{G}$ is the infinitesimal generator. The first-order condition for investment is $V_K^F = 1 + \xi(I_t/K_t - \iota)$. As $\xi \to 0$, this reduces to Tobin's $q = 1$, making the firm indifferent over investment rates. Imposing the side constraint of output-variance minimization then uniquely determines the adjustment rule (16), as shown in the proof of Proposition 3.2.

# C Non-Stochastic Steady State

Setting $\sigma_A = \sigma_F = 0$, $f_t = 0$, and all time derivatives to zero, the steady state satisfies (17)–(20). Existence follows from the intermediate value theorem applied to the capital market-clearing condition, and uniqueness from the strict monotonicity of $\alpha\bar{A}(\bar{K}/\bar{L})^{\alpha-1}$ in $\bar{K}/\bar{L}$.

# D Lyapunov Stability: Detailed Verification

Let $V(\mathbf{x}) = (\ln K - \ln \bar{K})^2 + (\ln A - \ln \bar{A})^2 + f^2$. Applying Itô's formula:

$$\mathcal{G}V = 2(\ln K - \ln \bar{K}) \cdot \frac{\mu_K^{\text{drift}}}{K} + 2(\ln A - \ln \bar{A}) \cdot \left(\mu_A - \tfrac{1}{2}\sigma_A^2\right) + 2f \cdot (-\kappa f) + \frac{\sigma_A^2 A^2}{A^2} + \frac{\Sigma_{KK}}{K^2} + \sigma_F^2,$$

where $\mu_K^{\text{drift}}$ is the drift of $K_t$ from (21) and $\Sigma_{KK}$ is given in Appendix E. Near the steady state, the dominant negative term is $-2\kappa f^2$. A careful bounding argument (Meyn and Tweedie, 2012, Ch. 15) establishes $\mathcal{G}V \leq -cV + b$ globally for $c = \min(2\kappa, |\mu_A - \frac{1}{2}\sigma_A^2|) > 0$ and a finite constant $b$.

# E Fokker–Planck Diffusion Coefficients

From the buffer rule (21), the capital diffusion coefficients entering (28) are:

$$\Sigma_{KK} = \left(\frac{1-\alpha}{\alpha}\frac{K}{L}\right)^2 \left(\theta^2\sigma_A^2 A^2 + \eta^2\sigma_F^2\right), \tag{43}$$

$$\Sigma_{KA} = -\left(\frac{1-\alpha}{\alpha}\frac{K}{L}\right)\theta\,\sigma_A^2 A^2. \tag{44}$$

These expressions follow directly from substituting (22) and (23) into (21) and applying the Itô isometry.

# F Welfare Proof

We use a second-order perturbation around the deterministic steady state. For CRRA utility with $\gamma \neq 1$, the welfare correction from output variance is $-(\gamma-1)\text{Var}(\ln C_t)/(2\rho)$ per unit time (with the sign chosen so that higher variance reduces welfare when $\gamma > 1$).

In the SAGE, consumption variance tracks output variance $\sigma_A^2\bar{Y}^2$ (liquidity shocks are buffered). In the un-buffered equilibrium, an additional liquidity-induced variance $(1-\alpha)^2\eta^2\sigma_F^2\bar{Y}^2/(2\kappa)$ enters. The welfare gain from buffering—the difference in welfare corrections—is:

$$\text{Gain} = \frac{(\gamma-1)(1-\alpha)^2\eta^2\sigma_F^2\bar{Y}^2}{2\rho \cdot 2\kappa}.$$

The per-unit-time adjustment cost of the buffer rule is approximately $\xi\iota^2(\sigma_F^2/2\kappa)\bar{K}$. Converting the gain to consumption-equivalent units and equating gain to cost yields equations (32) and (33).

# G Monetary Policy Rule

When $\xi > 0$, the capital adjustment rule (16) is implemented only partially each period. A central bank that observes $f_t$ can accelerate the buffer adjustment by setting the policy rate according to the augmented Taylor rule (38) with $\phi_f = (1-\alpha)\eta\kappa/(1+\phi_\pi)$. This choice exactly offsets the residual liquidity effect on output that arises when $\xi > 0$.

# H Numerical Methods

## Fokker–Planck Discretization

We use a uniform grid with $N_K \times N_A \times N_f = 50 \times 50 \times 30$ points. The state domain is $[\bar{K}e^{-5\sigma_K^*},\ \bar{K}e^{5\sigma_K^*}] \times [\bar{A}e^{-5\sigma_A\sqrt{T}},\ \bar{A}e^{5\sigma_A\sqrt{T}}] \times [-5\sigma_F/\sqrt{2\kappa},\ 5\sigma_F/\sqrt{2\kappa}]$, where $\sigma_K^* = \frac{1-\alpha}{\alpha}\sqrt{\theta^2\sigma_A^2 + \eta^2\sigma_F^2/(2\kappa)}$ is the approximate standard deviation of $\ln K_t$ derived from (43). The advection–diffusion equation is discretized using the Chang–Cooper scheme, which guarantees non-negativity and mass conservation.

## Euler–Maruyama Convergence

The Euler–Maruyama scheme for the SDE system achieves strong convergence of order $\mathcal{O}(\sqrt{\Delta t})$ and weak convergence of order $\mathcal{O}(\Delta t)$ (Øksendal, 2013, Ch. 9). With $\Delta t = 0.01$, the weak-convergence error is of order $10^{-4}$, well below the quantities of interest.

# The End